calcium-sensitive dye, they assessed how Ca^{2+} is distributed in the egg before and during fertilization. They found that Ca^{2+} spread across the egg in a wave that correlated with the appearance of the fertilization envelope.

Further studies demonstrated that the binding of a sperm to the egg activates a signal transduction pathway that triggers release of Ca^{2+} into the cytosol from the endoplasmic reticulum. The resulting increase in Ca^{2+} level causes cortical granules to fuse with the plasma membrane. A cortical reaction triggered by Ca^{2+} also occurs in vertebrates such as fishes and mammals.

Egg Activation

Fertilization initiates and speeds up metabolic reactions that bring about the onset of embryonic development, "activating" the egg. For example, cellular respiration and protein synthesis in the egg markedly accelerate following the entry of the sperm nucleus. Soon thereafter, the egg and sperm nuclei fully fuse, and cycles of DNA synthesis and cell division begin.

What triggers activation of the egg? A major clue came from experiments demonstrating that the unfertilized eggs of sea urchins and many other species can be activated by an injection of Ca^{2+}. Based on this discovery, researchers concluded that the rise in Ca^{2+} concentration that causes the cortical reaction also causes egg activation. Further experiments revealed that artificial activation is possible even if the nucleus has been removed from the egg. This further finding indicates that the proteins and mRNAs required for activation are already present in the cytoplasm of the unfertilized egg.

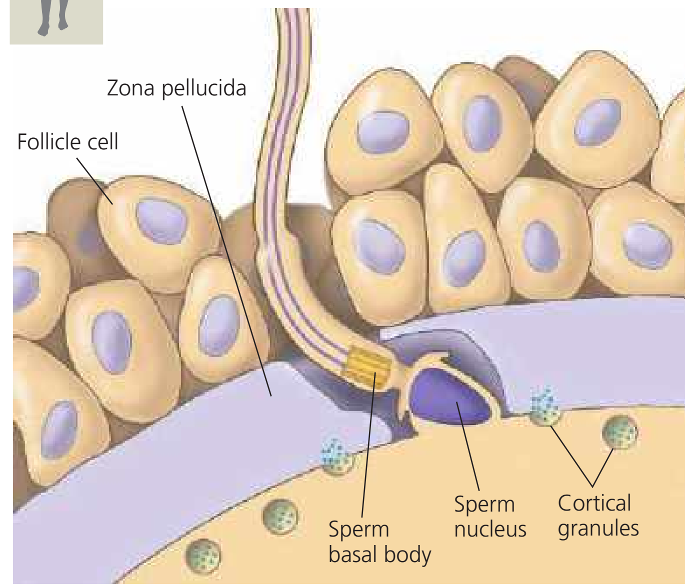
Not until about 20 minutes after the sperm nucleus enters the sea urchin egg do the sperm and egg nuclei fuse. DNA synthesis is then initiated. The first cell division, which occurs after about 90 minutes, marks the end of the fertilization stage.

Fertilization in other species shares many features with the process in sea urchins. However, there are differences, such as the stage of meiosis the egg has reached by the time it is fertilized. Sea urchin eggs have already completed meiosis when they are released from the female. In many other species, eggs are arrested at a specific stage of meiosis and do not complete the meiotic divisions until a sperm head enters. Human eggs, for example, are arrested at metaphase of meiosis II until sperm entry (see Figure 46.11).

Fertilization in Mammals

Unlike sea urchins and most other marine invertebrates, terrestrial animals, including mammals, fertilize their eggs internally. Support cells of the developing follicle surround the mammalian egg before and after ovulation. As shown in **Figure 47.4**, a sperm must travel through this

▼ **Figure 47.4 Fertilization in mammals.** The sperm shown here has traveled through the follicle cells and zona pellucida and has fused with the egg. The cortical reaction has begun, initiating events that ensure that only one sperm nucleus enters the egg.



layer of follicle cells before it reaches the **zona pellucida**, the extracellular matrix of the egg. There, the binding of a sperm to a sperm receptor induces an acrosomal reaction, facilitating sperm entry.

As in sea urchins, sperm binding triggers a cortical reaction, the release of enzymes from cortical granules to the outside of the cell. These enzymes catalyze changes in the zona pellucida, which then acts as the slow block to polyspermy. (No fast block to polyspermy is known in mammals.)

Overall, the process of fertilization is much slower in mammals than in sea urchins: The first cell division occurs within 12–36 hours after sperm binding in mammals, compared with about 1.5 hours in sea urchins. This cell division marks the end of fertilization and the beginning of the next stage of development, cleavage.

Cleavage

The single nucleus in a newly fertilized egg has too little DNA to produce the amount of mRNA required to meet the cell's need for new proteins. Instead, initial development is carried out by mRNA and proteins deposited in the egg during oogenesis. There is still a need, however, to restore a balance between the cell's size and its DNA content. The process that addresses this challenge is **cleavage**, a series of rapid cell divisions during early development **(Figure 47.5)**.